

Vincenzo Ruzzi

Curriculum Vitae

* 27 January 1995

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® Vincenzo Ruzzi



Education and Career

- Oct. 2026 – **PostDoctoral Research Fellow**, *SoMeX Lab, University of Vienna, Austria, project NEFCOEG - Sedimenting Colloids, funded by FFG*, Supervisor: Prof. Roberto Cerbino.
- Mar. 2028
- Sep. 2023 – **PostDoctoral Research Fellow**, *Soft Matter team, Laboratoire Charles Coulomb, CNRS, Université Montpellier*, supervisors: Laurence Ramos and Luca Cipelletti.
- Aug. 2026
- May 2020 – **PhD in Industrial Chemistry and Chemical Engineering**, *cum Laude, Soft Matter Lab, Dipartimento di Chimica, Materiali e Ingegneria Chimica "Giulio Natta", Politecnico di Milano*, supervisor: Roberto Piazza.
- Jul. 2023
- PhD Thesis (english): Optical Investigation of nonequilibrium properties of Soft Solids.
- Feb. 2020 – **Research Fellow Intern**, *Dipartimento di Chimica, Materiali e Ingegneria Chimica "Giulio Natta", Politecnico di Milano*, supervisor: Roberto Piazza.
- Apr. 2020
- Sep. 2017 – **Master of Science (Laurea Magistrale) in Nuclear Engineering**, *Politecnico di Milano*, mark: 110/110 *cum Laude*, supervisor: Roberto Piazza.
- Dec. 2019
- Master Thesis (english): Thermal Forces in 2D Colloids.
- Sep. 2014 – **Bachelor of Science (Laurea Triennale) in Engineering Physics**, *Politecnico di Milano*, mark: 107/110.
- Jul. 2017
- Bachelor Thesis (italian): Superconduttori: proprietà di trasporto e magnetiche.

Research

- **Scattering and optical techniques for the investigation of soft condensed matter:**
 - Static and Dynamic Light Scattering (SLS and DLS), Photon Correlation Imaging (PCI);
 - X-ray scattering: Small Angle X-ray Scattering (SAXS) and X-ray Photon Correlation Spectroscopy (XPCS);
 - Optical Microscopy and Differential Dynamic Microscopy (DDM);
 - Thermal Lensing (TL);
- **Opto-thermal stimulation of colloidal suspensions and soft solids;**
- **Microscopic dynamics of multidisperse colloidal and polymer solutions;**
- **Phase behavior, dynamics and microrheology of physical gels;**
- **Enzymatic gelation and degradation of biopolymer solutions and gels;**

During my PhD, I mainly focused on the optothermal response of soft materials through optical techniques, ranging from quasi-2D colloidal suspensions to physical gels. I have built an optical microscope to study the crystallization of colloidal suspensions geometrically confined in a quasi-2D geometry, induced by the action of an optically-generated thermal gradient. I have applied the same optothermal stimulation method to thermoresponsive systems, ranging from microgel suspensions to physical gels of biological interest using other optical techniques, such as TL and advanced multispeckle DLS methods.

In my PostDoc in Montpellier, I study the enzymatic degradation of hemicellulose polymer solutions and gels and the propagation of enzymes through these substrates, in the framework of the project CATCHing enzyMes in action (CATCHY), financed by several national grants (LabEX Numev, EXPLORAE, ANR). I have designed and built a novel optical setup that measures simultaneously the propagation of fluorescent enzymes and the microscopic dynamics of the substrate by combining Fluorescence and PCI, and developed data analysis tools to connect the spatio-temporal evolution of the degree of degradation of the substrates to the propagation mechanisms of the enzymes. This project has a strong interdisciplinary character, since it is carried out in collaboration with the Toulouse Biotechnology Institute (TBI) and the Institut Agro of Montpellier, bridging physics, biochemistry and enzymology.

Besides these main research lines, I have developed a general interest in statistical optics applied to soft condensed matter, colloids diffusion in simple and complex environments, and X-ray induced synthesis of nanoparticles. I have been Principal Investigator (PI) or co-investigator of five experiments in european synchrotron facilities for SAXS and XPCS experiments:

- **ESRF**, Grenoble: four experiments, one as PI, 288 hours beamtime;
- **SOLEIL**, Paris: one experiment as PI, 48 hours beamtime;

I am currently involved in collaborations developed over the past years with D. Truzzolillo (CRNS, Montpellier), D. Parisi (University of Groningen), I. Morfin (LIPHY and ESRF, Grenoble) and E. Secchi (ETH, Zurich).

Skills

- **Experimental Skills:** I am an experimental soft matter physicist with a training as engineer encompassing optics, electronics, condensed matter physics and chemistry. My main quality consists in the ability to design custom experiments (often non-conventional) and data analysis tools for a targeted system/question in the world of soft matter, using mainly optical techniques.
- **Programming Skills:** Python, C, Matlab, ImageJ, FlexPDE, Mathematica. I developed custom made codes in Python and Matlab for most of the experiments and data analysis I have carried out. I also performed finite element simulations to solve partial differential equations for modeling purposes (FlexPDE).
- **Personal Skills:** I had the opportunity to work with researchers from all Europe in the framework of interdisciplinary projects (CATCHY) and in large-scale facilities. These experiences allowed me to establish collaborative and wide-spectrum research networks, in which I could disseminate my work to a broad and diverse audience. During my PostDoc in Montpellier, I was part of the organizing committee of the L2C Non-Permanent Day, a social event gathering PhD students and postdoctoral researchers of the department from different fields of Physics, with the aim of presenting their research to a diversified environment, while establishing a friendly network in the

community.

Publications

I have published one preprint and nine papers in peer reviewed journals. I am either first or corresponding author in five of them.

- 2026 **Simultaneous measurements of enzyme propagation and biopolymer mobility elucidate the directional degradation of a dense biopolymer matrix**, V. Ruzzi, A. Bouchoux, C. A. Assor, D. J. Bigot, S. Menzeh, M. Petermann, L. Leclercq, H. Cottet, C. Montanier, C. Dumon, L. Cipelletti, L. Ramos, *arXiv preprint* arXiv:2609.03792 (2026)
- Decoupling structure and elasticity in colloidal gels under isotropic compression**, M. Milani, E. Cavalletti, V. Ruzzi, A. Martinelli, P. Dieudonné-George, C. Ligoure, T. Phou, L. Cipelletti, L. Ramos, *Physical Review Letters* (2026), **136**.3: 0382011.
- 2024 **Short-time self-diffusion in binary colloidal suspensions***, V. Ruzzi, S. Buzzaccaro, P. Moretti, R. Piazza, *The Journal of Chemical Physics* (2024), **161**, 121102.
*Promoted as Editor's Pick.
- 2023 **Optothermal crystallization of hard spheres in an effective bidimensional geometry**, V. Ruzzi, J. Baglioni, R. Piazza, *The Journal of Chemical Physics* (2023), **159**.15, 154904.
- Characterization of the physical properties of the thermoresponsive block-copolymer PAGB2000 and numerical assessment of its potentialities in Forward Osmosis desalination**, I. M. Carraretto, V. Ruzzi, F. Lodigiani, R. Colciaghi, R. Simonetti, S. Buzzaccaro, L. Molinaroli, L. P. M. Colombo, R. Piazza and G. Manzolini, *Polymer Testing* (2023), 108238.
- Heterogeneity governs 3D-cultures of clinically relevant microbial communities**, D. P. Pacheco, F. Bertoglio, C. Butnarusu, N. Suarez Vargas, G. Guagliano, A. Ziccarelli, F. Briatico-Vangosa, V. Ruzzi, S. Buzzaccaro, R. Piazza, S. van Uden, E. Crotti, S. Visentin, L. Visai, P. Petrini, *Advanced Functional Materials* (2023), 2306116.
- A light scattering investigation of enzymatic gelation in self-assembling peptides**, S. Buzzaccaro, V. Ruzzi, F. Gelain, R. Piazza, *Gels*, **9**.4 (2023), 347.
- Thermal lens measurements of thermal expansivity in thermosensitive polymer solutions**, V. Ruzzi, S. Buzzaccaro and R. Piazza, *Polymers*, **15** (2023), 1283.
- 2022 **Microrheology of a thermosensitive gelling polymer for cell culture**, S. Buzzaccaro, V. Ruzzi, T. Faleo, R. Piazza, *The Journal of Chemical Physics*, **157** (2022), 174901.
- Unraveling gelation kinetics, arrested dynamics and relaxation events in filamentous colloids by photon correlation imaging**, M. Uselli[†], V. Ruzzi[†], S. Buzzaccaro, G. Nyström, R. Piazza, R. Mezzenga, *Soft Matter*, **18** (2022), 5632.
[†]Equal contribution.

Conferences and schools

During my career, I have attended eleven national and international conferences and I have given ten oral presentations.

- 2026 **Catching enzymes in action throughout the degradation of hemicellulose biopolymers**, *oral presentation*, V. Ruzzi, C.A. Assor, C. Dumon, C. Montanier, A. Bouchoux, L. Cipelletti and L. Ramos.
AERC 2026, Krakow, Poland.
- Catching enzymes in action throughout the degradation of hemicellulose biopolymers**, *oral presentation*, V. Ruzzi, C.A. Assor, C. Dumon, C. Montanier, A. Bouchoux, L. Cipelletti and L. Ramos.
Italian Soft Days, 7th edition, Naples, Italy.
- 2025 **Enzyme propagation and substrate microscopic dynamics throughout the enzymatic degradation of lignocellulosic polymers**, *oral presentation*, V. Ruzzi, C.A. Assor, C. Dumon, C. Montanier, A. Bouchoux, L. Cipelletti and L. Ramos.
39th ECIS conference, Bristol, UK.
- Radiolysis-induced synthesis of mercury nanodroplets from an organometallic preservative**, *oral presentation*, V. Ruzzi, A. Martinelli, A. Hubley, I. Morfin and L. Ramos.
GdR Slamm, Obernai, France.
- 2024 **Enzymatic degradation of biopolymers: a combined light scattering-fluorescence study**, *oral presentation*, V. Ruzzi, C.A. Assor, C. Dumon, C. Montanier, A. Bouchoux, L. Cipelletti and L. Ramos.
GdR Slamm, Le Croisic, France.
- Enzymatic degradation of biopolymers: a combined light scattering-fluorescence study**, *poster*, V. Ruzzi, C.A. Assor, C. Dumon, A. Bouchoux, L. Cipelletti and L. Ramos.
Italian Soft Days, 6th edition, Florence, Italy.
- Enzymatic degradation of biopolymers: a combined light scattering-fluorescence study**, *oral presentation*, V. Ruzzi, C.A. Assor, C. Dumon, A. Bouchoux, L. Cipelletti and L. Ramos.
20th SoftComp Annual Meeting, Lyon, France.
- 2023 **Nucleation and growth of an optothermally-generated 2D colloidal crystal**, *oral presentation*, V. Ruzzi, J. Baglioni and R. Piazza.
1st UniMi Complex System Community Day, Milan, Italy.
- Optothermal heating effects on the structure and dynamics of a soft disordered solid**, *oral presentation*, V. Ruzzi, S. Buzzaccaro and R. Piazza.
15th International Meeting on Thermodiffusion, Tarragona, Spain.
- 2022 **Phase behavior, microscopic dynamics and microrheological properties of a thermosensitive gel-forming polymer**, *poster*, V. Ruzzi, S. Buzzaccaro and R. Piazza.
Italian Soft Days, 5th edition, Bari, Italy.

Nucleation and growth of an optothermally-generated 2D colloidal crystal, oral presentation, V. Ruzzi, S. Buzzaccaro and R. Piazza.
Italian Soft Days, 5th edition, Bari, Italy.

Phase behavior, microscopic dynamics and microrheological properties of a thermosensitive gel-forming polymer, oral presentation, V. Ruzzi, S. Buzzaccaro and R. Piazza.
36th ECIS conference, Chania, Crete, Greece.

Jul. 2021 **Advanced School in Soft Condensed Matter ‘Solutions in the Summer’**, IOP, Liquids and Complex Fluids Group.

Memberships of Scientific Society

- **European Colloid and Interface Society (ECIS)**
- **European Society of Rheology (ESR)**

Co-supervision and tutoring activity

- 2025 – in progress **Rim Dannaoui**, *PhD in Physics*, Dynamic Monitoring of the Encapsulation and Release of Drugs in a Bio-sourced Gel.
Supervisors: Dr. Laurence Ramos and Prof. Luca Cipelletti.
- 2023 **Matteo Bessega**, *Master of Science in Nuclear Engineering*, Dual Beam Setup for the Measurement of Thermal Lensing in Complex Fluids.
Supervisor: Prof. Roberto Piazza.
- Rafael Gervasone**, *Master of Science in Materials Engineering and Nanotechnology*, Optothermal Effects in Soft Disordered Solids.
Supervisor: Prof. Stefano Buzzaccaro.
- 2022 **Filippo Faedi**, *Master of Science in Chemical Engineering*, Brownian dynamics and gravity settling of binary colloidal mixtures.
Supervisor: Prof. Stefano Buzzaccaro.
- 2020 **Jacopo Baglioni**, *Master of Science in Nuclear Engineering*, Optothermal Manipulation of 2D Complex Fluids and Soft Solids.
Supervisor: Prof. Roberto Piazza.

Teaching

- 2021 – 2022 **Tutoring activity in Thermodynamics**, Course: Fondamenti di Fisica Tecnica per l'Ingegneria Chimica, Prof. Francesco Cellesi.
- 2021 **Tutoring activity in Physics I**, Course: Fisica I, Prof. Andrea Crespi.

Languages

Italian, native speaker.
English, fluent.
French, intermediate.

Vienna, 2 October 2026